# SUPERSTORE BUSINESS INTELLIGENCE REPORT

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# SUPERSTORE BUSINESS INTELLIGENCE REPORT

Analysis Period: March 2014 – April 2017

## EXECUTIVE SUMMARY

This report presents the findings of the Business Intelligence Analysis conducted on the Superstore dataset from 2014 to 2017. The sales, product, and regional performance metrics were evaluated, as well as the customer behavior, shipping efficiency, and the impact of discounts on the business. The analysis revealed the following insights:

- The company's revenue grew by 51.41%, while total profits increased by 88.60%. Also, profit margins increased from 10.23% in 2014 to 12.47% in 2017, indicating a substantial growth in revenue and profitability.
- The West region is the highest performing region while the South region sees the lowest sales, but it has high profitability with a profit margin of 11.93%.
- Technology and office supplies are the most profitable categories, each generating approximately 17% in profit margins.
- Discounts more than 20% contribute to losses in profit, as customers are buying products with discounts less than 20%.
- During the analysis period, a total of 793 customers made 5009 orders. This shows that customers were making repeat orders, indicating strong customer satisfaction.

## INTRODUCTION

Superstore contracted an analysis and use of its sales, profit, customer, and location data to build an executive dashboard that will enable management to:

- Monitor sales performance
- Monitor profitability
- Monitor customer behavior
- Monitor regional performance
- Monitor the effect of discounts on sales and profit
- Make strategic decisions

This report summarizes key findings from March 2014 to April 2017 and provides actionable recommendations to support efficient business decision-making. This analysis was performed using Power BI to evaluate key business metrics and uncover trends related to sales, profitability, customers, products, location, shipping methods, and discounting practices.
The primary goals of the analysis were to:

- Assess overall business performance.
- Identify high-performing and underperforming regions.
- Evaluate customer and product profitability.
- Understand the impact of discounts on sales and profits.
- Provide actionable recommendations to improve business outcomes.

## BUSINESS OBJECTIVES

The analysis aimed to answer the following questions:

- How has sales and profit performance changed over time?
- Which regions, states, and cities generate the highest revenue and profit?
- Which customer segments contribute the most to revenue?
- Which product categories perform best?
- How efficient are existing shipping methods?
- What impact do discounts have on revenue and profitability?

## KEY PERFORMANCE INDICATORS

The following Key Performance Indicators (KPIs) were calculated to aid analysis:

*Table 1: KPIs, calculations, and results used during analysis*

| KPI | DAX Measure | Result |
|---|---|---|
| **Total Sales** | Total Sales = SUM(Superstore[Sales]) | $2.3M |
| **Total Profit** | Total Profit = SUM(Superstore[Profit]) | $286.40K |
| **Profit Margin** | Profit Margin = DIVIDE([Total Profit],[Total Sales],0) | 12.47% |
| **Total Orders** | Total Orders = DISTINCTCOUNT(Superstore[Order ID]) | 5009 |
| **Average Sales** | Average Sales = AVERAGE(Superstore[Sales]) | $229.86 |
| **Total Customers** | Total Customers = DISTINCTCOUNT(Superstore[Customer ID]) | 793 |
| **Total Products** | Total Products = DISTINCTCOUNT(Superstore[Product ID]) | 1862 |
| **Overall Average Shipping Time** | Overall Avg. Shipping Days = AVERAGE(Superstore[Days To Ship]) | 4 days |
| **Average Standard Shipping Time** | Avg. Standard Shipping Days = CALCULATE(AVERAGE(Superstore[Days To Ship]), Superstore[Ship Mode] = "Standard Class") | 5 days |

# KEY FINDINGS

## SALES PERFORMANCE

**What is the overall sales performance of the company? How has sales and profit changed over time?**

- Total sales increased from $484,247.47 in 2014 to $733,215.05 in 2017, representing a growth of 51.41% over the four-year period.
- Total profit grew by 88.60%, faster than total sales, growing from $49,543.85 in 2014 to $93,439.36 in 2017.
- In addition, the company's profit margin also increased from 10.23% in 2014 to 12.47% in 2017. This indicates that the business is profitable; however, profitability decreased from 13.43% in 2016 to 12.47% in 2017.

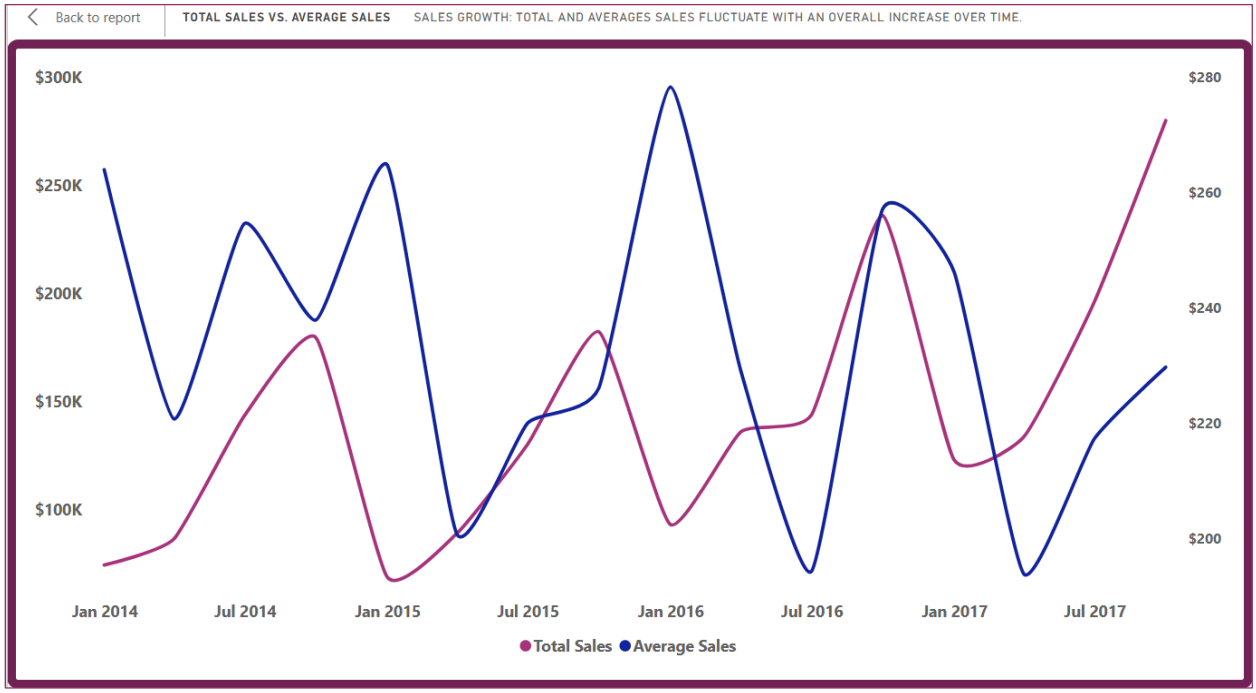



*Figure 1: Total sales vs. average sales: A 51.41% increase in sales from 2014 to 2017.*

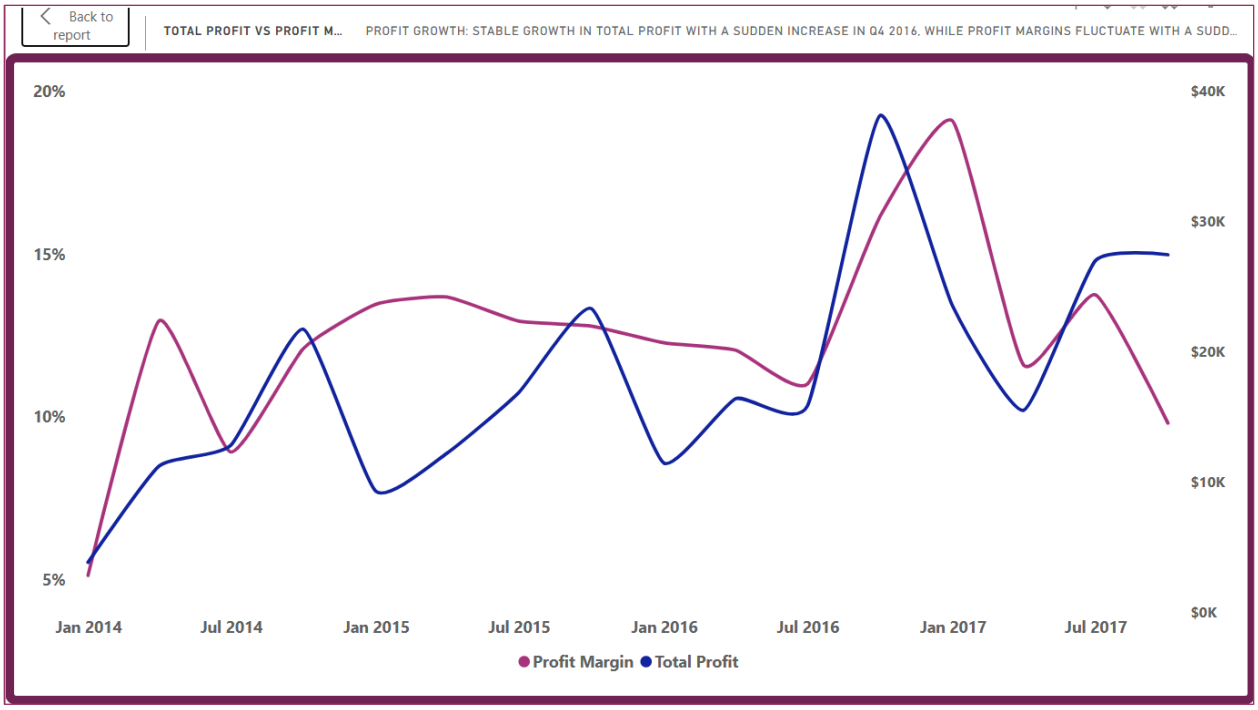



*Figure 2: Total profit vs. profit margin: A profit margin of 12.47% and an increase in total profits of 88.60%.*

## LOCATION PERFORMANCE

### Which regions, states, and cities generate the highest revenue and profit?

- The West region generated the highest sales and profit. It is also the most profitable region with a profit margin of 14.94%.
- Although the South region generated the lowest sales revenue, it achieved a profit margin of 11.93%, significantly higher than the Central region's 7.92% profit margin.
- The highest-performing locations are California State and New York City. In contrast, Texas and Philadelphia generated a total loss of -$25,729.32 and -$13,837.84, respectively.

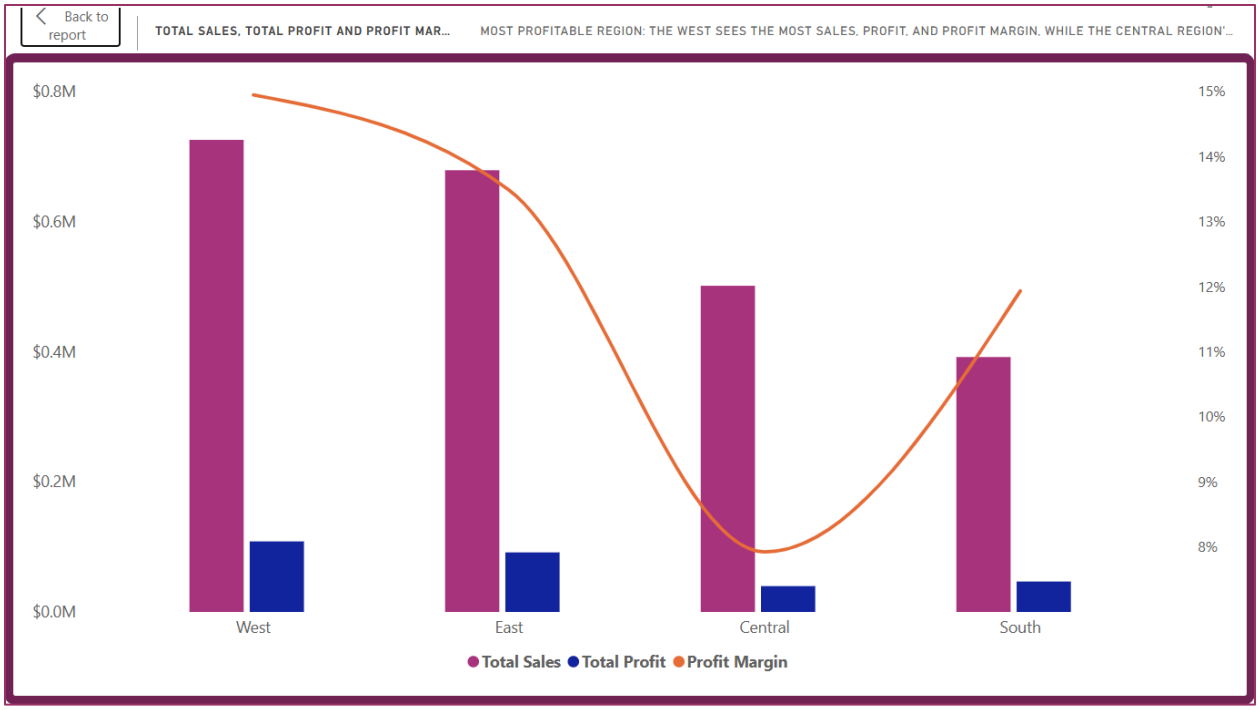



*Figure 3: Sales and Profit by Region: The west region generates the highest sales, profit and is the most profitable region.*

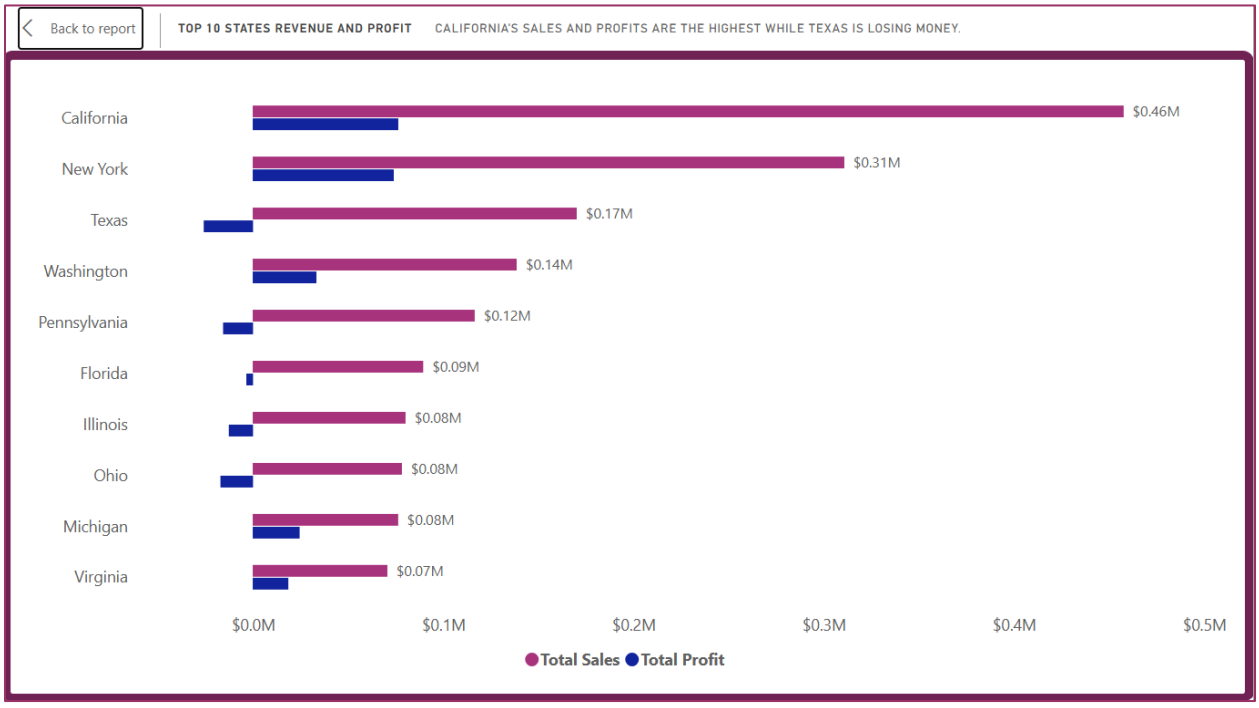



*Figure 4: Top 10 States: California generates the highest revenue while Texas makes a loss.*

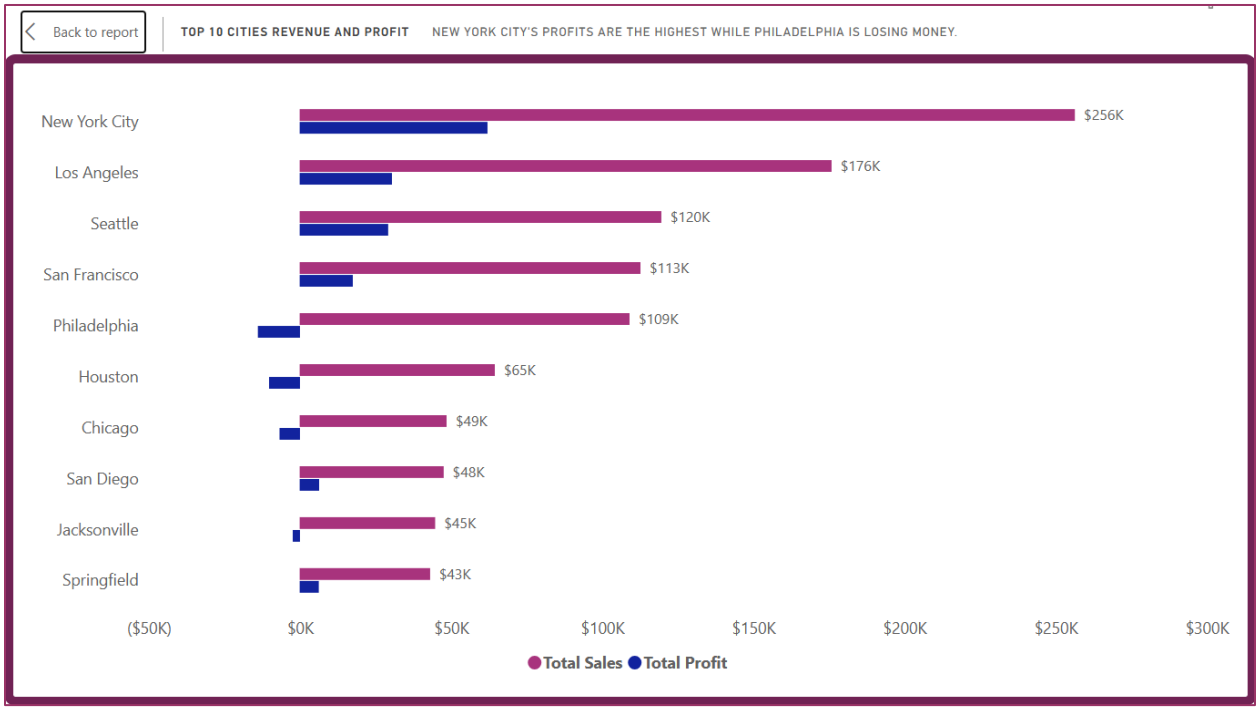



*Figure 5: Top 10 Cities: New York City is the highest selling city while Philadelphia loses money.*

## CUSTOMER SEGMENT ANALYSIS

### Which customer segments contribute the most to revenue?

- Superstores served a total of 793 customers who made approximately 5,000 orders. This suggests that customers are making repeat orders, demonstrating substantial customer satisfaction.
- The Consumer segment is the largest customer segment, contributing 44.95% of Superstore's total revenue. However, Corporate and Home Office customers have higher profit margins at 13.03% and 14.03%, respectively. This indicates that these two customer segments are more profitable than the Consumers segment.

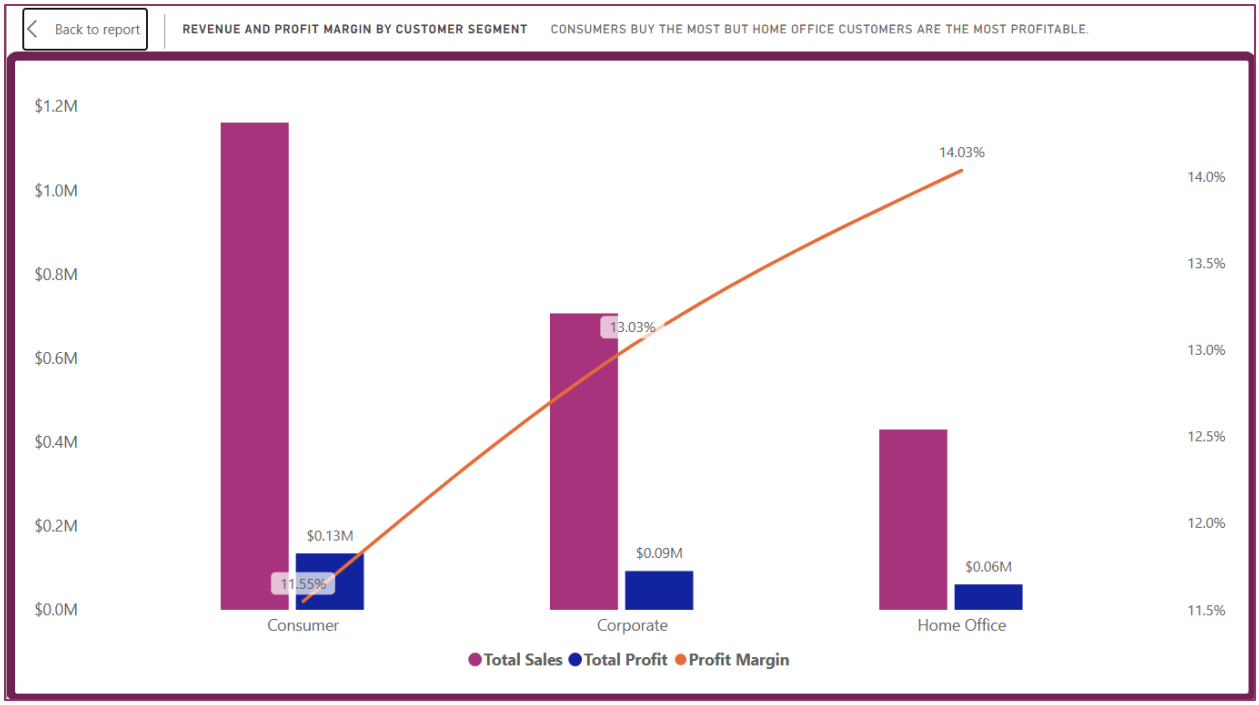



*Figure 6: Sales vs. Profit by customer segment: Home Office generates the lowest sales, yet has the highest profitability.*

## PRODUCT ANALYSIS

### Which product categories perform best? Which products are the most profitable?

- Technology and Office Supplies are the company's most profitable product categories, with a profit margin of approximately 17% each. Although furniture contributes a substantial amount of revenue, its profit margin is merely 2.49%, resulting in less profitability than the other two categories.
- Phones are the highest selling subcategory, generating a total revenue of $330,007.10. However, copiers generate more profit ($55,617.88) than phones ($44,516.16).

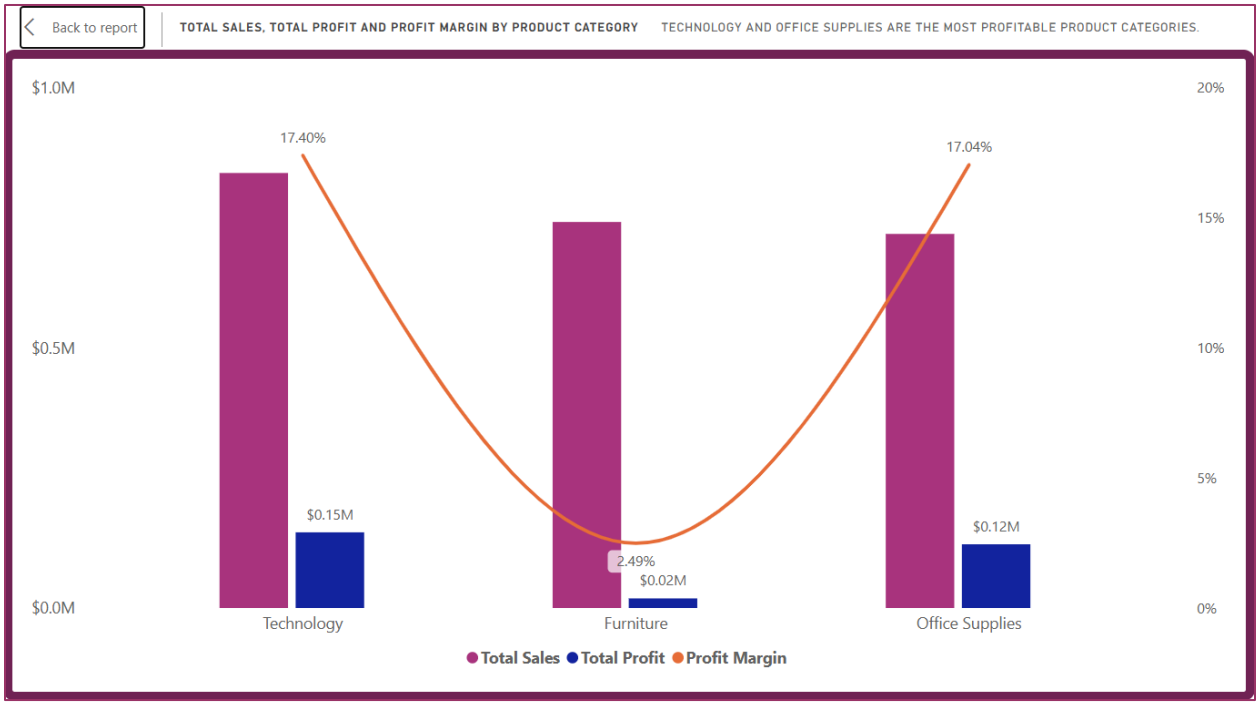



*Figure 7: Sales and profit: Furniture generates substantial revenue, but it has the lowest profit margin.*

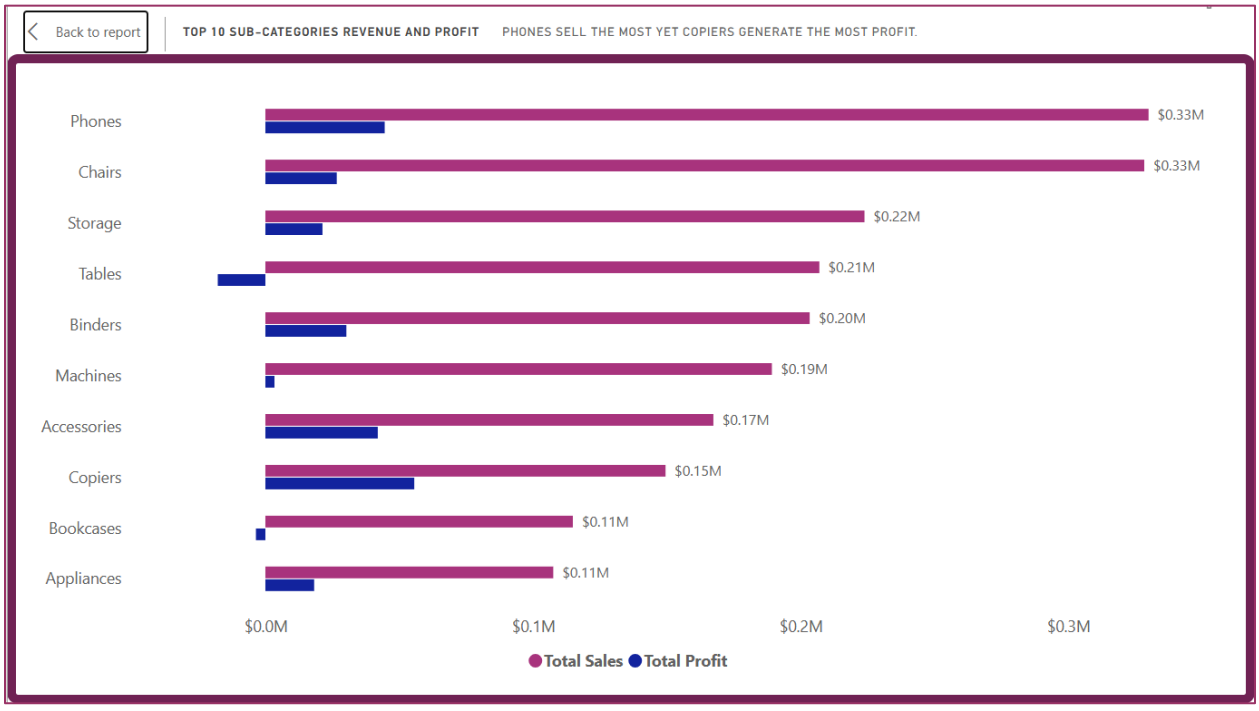



*Figure 8: Top 10 sub-categories: Phones generate the highest revenue, but copiers generated the highest total profit.*

## SHIPPING ANALYSIS

**Which shipping mode do customers prefer? How efficient are existing shipping methods?**

- Standard shipping accounts for 59.12% of total revenue, suggesting that it is the most popular shipping mode among customers.
- Overall average shipping time for orders is 4 days, but standard shipping takes 5 days on average. This is slower than the shipping time for normal orders.

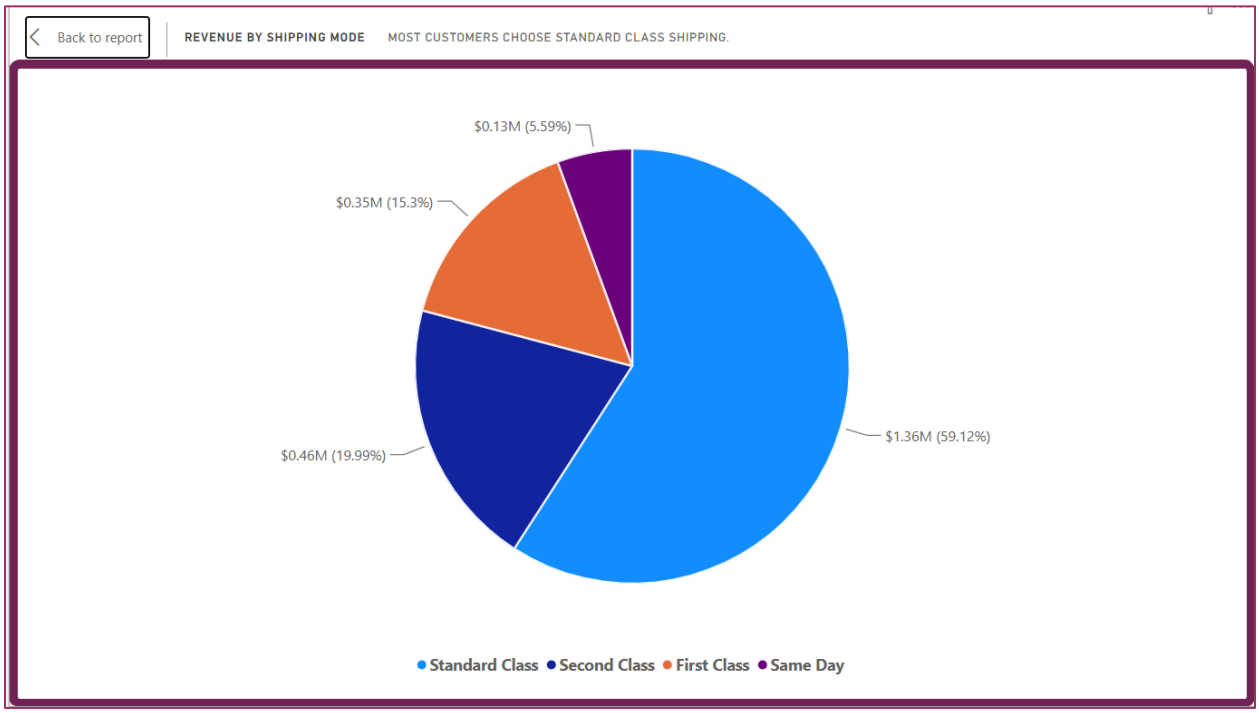



*Figure 9: Standard class shipping is the most popular among customers.*

## DISCOUNT ANALYSIS

### What impact do discounts have on revenue and profitability?

- Products with 0% discount generated the highest revenue and profit with a profit margin of 29.50%. Meanwhile, items with 50% or more discount had the lowest sales and generated a loss with a profit margin of -78.82%.
- Products with 10% to 20% discounts generated the second highest sales and profit with a profit margin of 11.91%.
- This shows that customers bought products with less than 20% discount more than items with more than 30% discount.

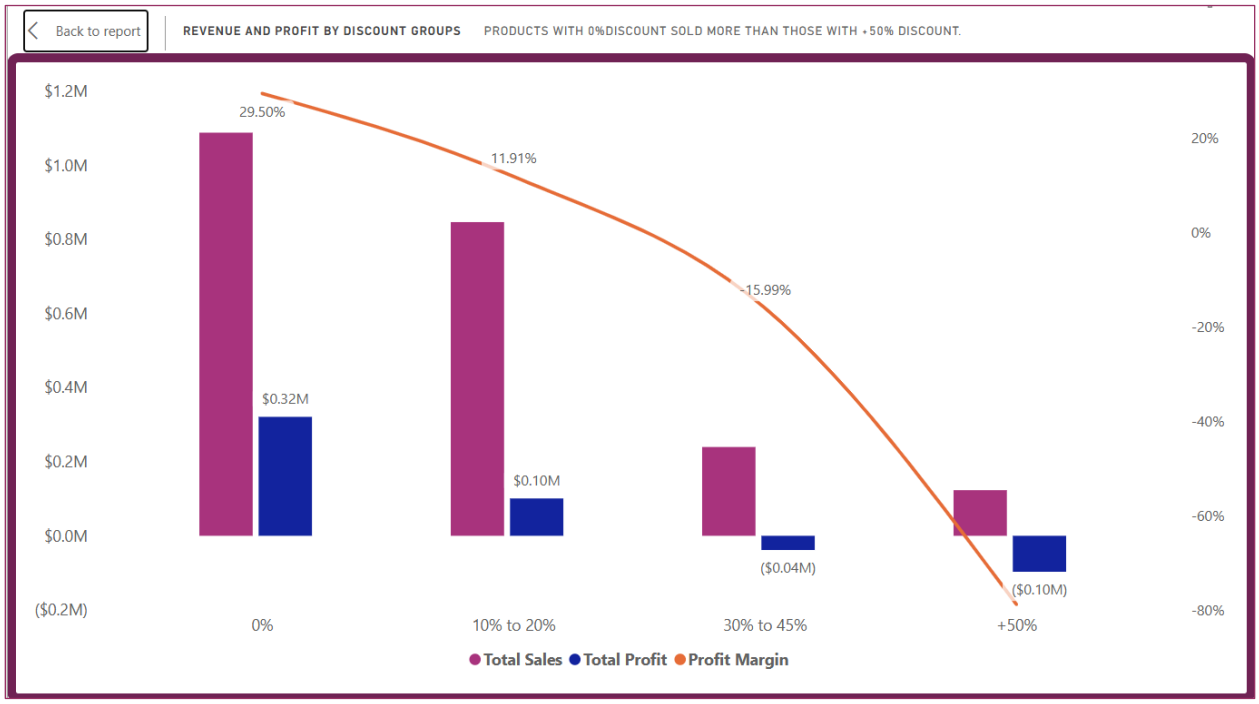



*Figure 10: Items with 0% discount are highly profitable with a margin of 29.50% while products with 30% or above discount do not sell well.*

## RECOMMENDATIONS

**What recommendations should management implement to improve business performance?**

Superstore should implement the following actions to improve business performance:

1. Investigate the decline in profit margin between 2016 and 2017.
2. The furniture category generates substantial revenue but has low profitability. Therefore, pricing and discount practices for this category should be assessed.
3. The loss in low-performing areas like Texas and Philadelphia should be investigated.
4. Corporate and Home Office customers should be retained through rewards and loyalty programs.
5. Ensure inventory is always available for high-performing product categories, and develop marketing campaigns to draw attention to low-performing products.
6. Reduce standard class shipping time from 5 days to 4 days or less to improve customer satisfaction.
7. Discount rates should remain below 20%. Discounting products above 30% should be approved by higher executives. Strict protocols should be established to ensure discounts above 30% on products are strictly regulated, since these types of discounts have a detrimental effect on sales and profits.
8. The dashboard should be monitored regularly to spot issues early. A review of the data should be conducted every quarter to discuss the impact of implementing these recommendations.

## ASSUMPTIONS

The following assumptions were made during the analysis:

- It was assumed that all the data provided was accurate, complete, and representative of actual business operations.
- The Sales, Profit, and Discounts were assumed to have been captured correctly.
- Repeat orders were assumed to indicate customer satisfaction with the company.
- The analysis assumes that discounts influence sales, profit, and profitability.

## LIMITATIONS

- The dataset covers the period from 2014 to 2017. Therefore, the findings may not accurately reflect current company performance.
- Cost information (such as cost price, selling price, or inventory costs) was not included in the dataset, limiting the analysis of profitability.

## RISKS

a) The furniture category generates a substantial amount of revenue yet its profitability is low. This suggests that the company is spending money on items that are not profitable, which could lead to further losses.

b) Customers appear to spend more money on items with low discounts than products with high discounts. Therefore, increasing discounts above 30% may be risky as it may result in more losses.

c) Low-performing areas such as Texas and Philadelphia generated losses and reduce the company's overall profitability.

d) The decline in profit margin between 2016 and 2017 may be a warning sign, thereby requiring further investigation.

## OPPORTUNITIES

a) Superstore should invest in the Technology and Office Supplies categories to ensure adequate inventory availability to meet customer demands.

b) Expand the Corporate and Home Office customer segments since they produce higher profit margins the than Consumers segment.

c) Despite lower sales, the South region demonstrated stronger profitability than the Central region. Therefore, management should consider investing more money into the South region.

## FURTHER ANALYSIS

The following aspects of the dataset can be analyzed further to answer more business questions:

1. Sales and profit in each discount group vs. category
2. sales and profit in each discount group per region
3. Profitability by discount group
4. The impact of discounts, shipping costs, and product performance in low-performing cities, states, and regions should be investigated further.
5. The impact of pricing, discounts, product availability on the furniture category should be investigated.
6. The impact of delivery time on repeat purchases should be investigated further.

## CONCLUSION

The Superstore business demonstrated strong growth between 2014 and 2017, achieving a 51.41% increase in sales and an 88.60% increase in profit. The company benefits from strong regional performance in the West, profitable Technology and Office Supplies categories, and loyal customers who make repeat purchases.

However, the analysis also revealed multiple opportunities for improvement, including optimizing discount strategies, improving Furniture profitability, reducing losses in underperforming markets, and enhancing shipping efficiency. By implementing the recommendations outlined in this report, Superstore can strengthen profitability, improve operational performance, and position itself for sustainable long-term growth.

# DASHBOARD SCREENSHOTS

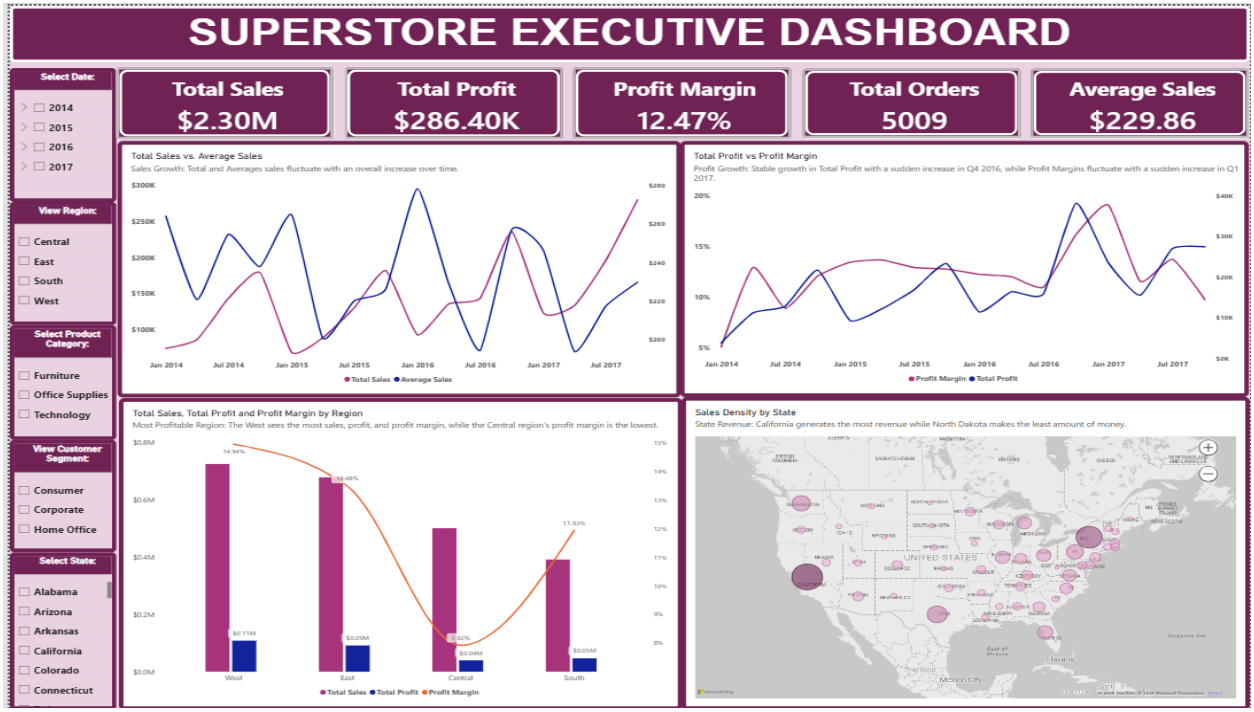



*Figure 11: Dashboard screenshot 1*

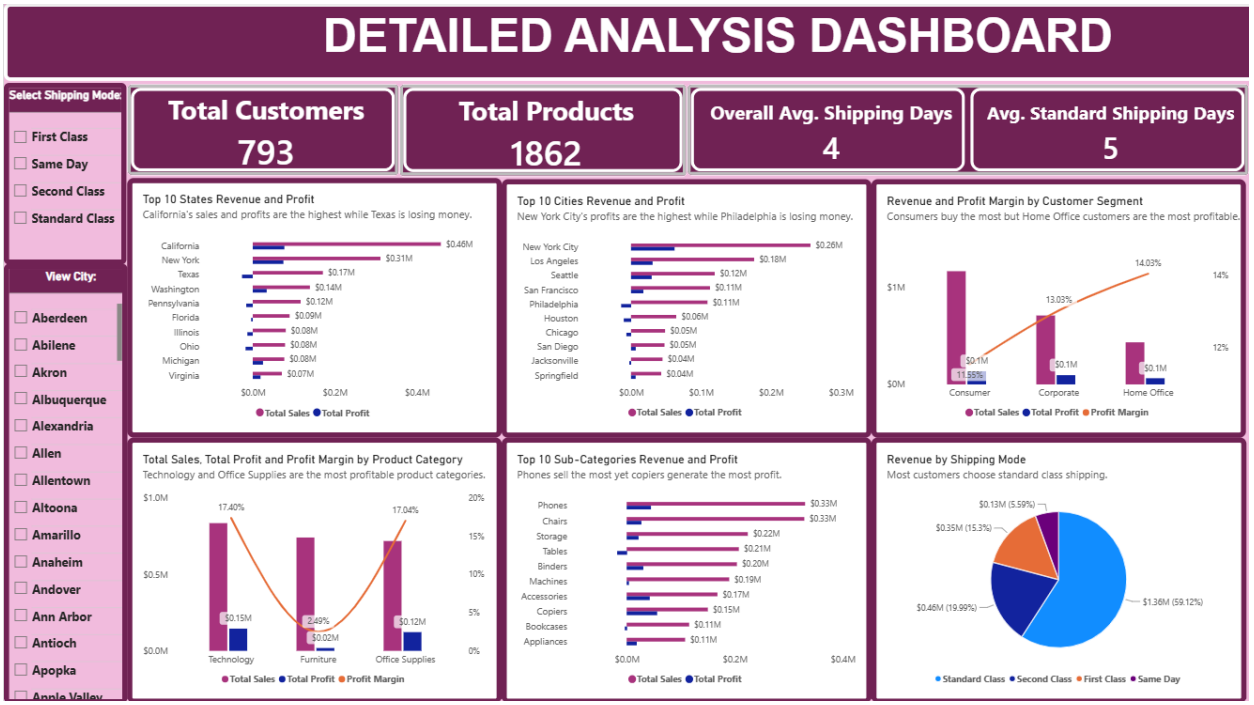



*Figure 12: Dashboard screenshot 2*

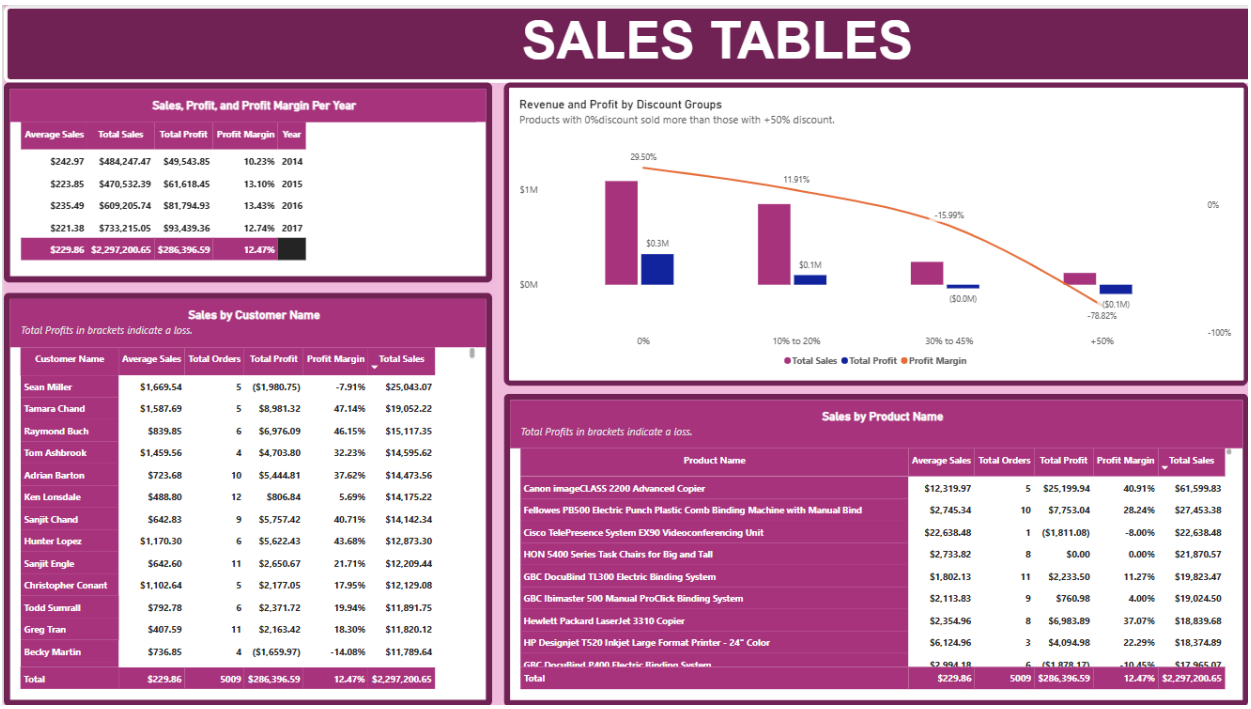



*Figure 13: Dashboard screenshot 3*